	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment 1
Branch	**Computer Science Engineering**	Programme	**B. Tech**
Course Name	Renewable Energy	Semester	**8TH**
Course Code	**OPL4001**	Academic Year	**2025-26**
Assignment- 1		**Topic:** Renewable Energy Principles	
Learning Level (LL)	**L1**: Remembering	**L3:** Applying	**L5**: Evaluating
	L2: Understanding	**L4**: Analysing	**L6**: Creating

Q's	Questions	COs	LL
1	Classify the energy resources based on commercial application with examples.	**CO1**	**L2**
Ans)			

Type of Energy Source	Definition	Description	Examples
Commercial Energy Sources	The energy sources that are available in the market for a definite price are known as commercial energy.	By far the most important form of commercial energy are electricity, coal and refined petroleum products. Commercial energy forms the basis of industrial, agriculture, transport, and commercial development in modern world. In industrialized countries, commercialized fuels are predominant sources not only for economic production, but also for many household tasks of general population.	Electricity, Lignite, Coal, oil etc.
Non-commercial Energy Sources	The energy sources that are not available in the market for a price are known as non-commercial energy sources.	Non-commercial energy sources include fuels such as firewood, cattle dung and agriculture waste, which are traditionally gathered, and not bought at a price used especially in households. These are also called as traditional fuels. Non-commercial energy is often ignored in energy accounting.	Firewood, agro waste from rural areas.

2	What do you mean by green power? Give some examples.	**CO1**	**L2**
Ans)	Green power refers to electricity generated from renewable energy sources that have minimal environmental impact and produce little or no greenhouse gas emissions. These energy sources are natural and self-replenishing and usually have a low- or zero-carbon footprint. Examples of Green Power • Solar energy • Wind energy • Hydropower • Biomass energy • Geothermal energy		
3	Explain the greenhouse effect. Mention some of the greenhouse gasses.	**CO1**	**L2**
Ans)	The greenhouse effect is the process by which heat is trapped in the earth's atmosphere by greenhouse gases, which allow sunlight to enter the atmosphere but prevent some of the heat from escaping back into space. This causes the temperature of the earth to increase. Greenhouse gases absorb infrared radiation and trap heat in the atmosphere, thereby warming the earth's surface. Greenhouse Gases		

	• Carbon dioxide (CO_2) • Methane (CH_4) • Nitrous oxide (N_2O) • Water vapour • Ozone		
4	Write the consequences of global warming.	**CO1**	**L2**
Ans)	The consequences of global warming include: • Increase in the average temperature of the earth. • Melting of glaciers and polar ice caps. • Rise in sea level which may cause flooding of coastal areas. • Changes in weather patterns and climate conditions. • Increase in the frequency of extreme events such as droughts, floods and storms. • Loss of biodiversity and disturbance to ecosystems.		
5	What are the different types of solar energy.	**CO2**	**L2**
Ans)	Solar power is energy from the sun that is converted into thermal or electrical energy. **Types of Solar energy:** • Direct • Indirect **Direct solar energy** • Solar heater • Solar pond • Solar cooker **Indirect solar energy** • Solar PV • Solar light • Solar cells		
6	What do you understand by Passive and Active solar energy? What is the difference between them?	**CO2**	**L2**
Ans)	Solar energy is radiant light and heat from the Sun that is harnessed using a range of ever-evolving technologies. It is an important source of renewable energy and its technologies are broadly characterized as either passive solar or active solar depending on how they capture and distribute solar energy or convert it into solar power. **Passive Solar Energy:** Passive solar systems use the design of buildings and materials to capture, store and distribute solar energy naturally without the use of mechanical devices. **Active Solar Energy:** Active solar systems use mechanical and electrical equipment such as pumps, fans and solar collectors to collect, store and distribute solar energy.		
7	What do you understand by beam radiation? What is the desirable value with respect to global radiation?	**CO2**	**L2**
Ans)	Beam radiation (Direct radiation) is the solar radiation that reaches the earth's surface directly from the sun without being scattered or diffused by the atmosphere. It travels in a straight line and is the main source of solar energy used in solar collectors and concentrators. Global radiation is the total solar radiation received on the earth's surface, which is the		

	sum of beam radiation and diffuse radiation. $$Ig = Ib + Id$$ Where: • **Ig** = Global radiation • **Ib** = Beam radiation • **Id** = Diffuse radiation For efficient solar energy utilization, a higher proportion of beam radiation with respect to global radiation is desirable, because beam radiation can be concentrated and used more effectively in solar energy systems.		
8	How water vapor controls the beam radiation.	**CO2**	**L2**
Ans)	When solar radiation passes through the earth's atmosphere, water vapour absorbs and scatters part of the solar radiation. Because of this absorption and scattering, the amount of direct beam radiation reaching the earth's surface decreases. Water vapour present in the atmosphere reduces the intensity of beam radiation by absorbing some of the solar energy and converting it into heat. This process also increases diffuse radiation, since some of the solar rays are scattered in different directions. Therefore, the amount of water vapour in the atmosphere controls the intensity of beam radiation reaching the earth's surface.		
9	Explain about solar constant and inclination angle.	**CO2**	**L2**
Ans)	**Solar Constant:** Solar constant is the rate at which energy is received from the sun on a unit area perpendicular to the rays of the sun at the mean distance of the earth from the sun. The solar constant is approximately 1370 W/m². **Inclination Angle:** The angle between the sun's ray and its projection on horizontal surface is known as inclination angle. It is the angle of incidence between the beam radiation and the normal to the plane surface or collector plate.		
10	What are the different renewable energy policies by 2025	**CO2**	**L2**
Ans)	The renewable energy policies aim at promoting clean energy development and improving energy security. Important policy strategies include: • Enhancing Energy Security • Harnessing Innovation • Modernizing the Power System • Enhancing Access to Affordable Energy These strategies help in promoting renewable energy adoption and sustainable energy development.		

Note:

1. Assignment carries a weightage of 10 **marks out of 100**
2. CO1 and CO2 are covered.

Course Outcomes	CO1	Understand different sources of energy and their effects in developing aware ness on impacts of global warming.
	CO2	Solar Thermal energy and Understanding how Solar radiation reaches the earth's surface.
	CO3	Solar Photo Voltaic system, its type and its integration.

	CO4	Hydro energy, types of hydroelectric plant its environmental impact.
	CO5	Tidal power, Power generation from tidal barrages and its environmental impact.
	CO6	Wind energy, different types of wind turbine. Power and energy from wind turbine and its environmental impact.